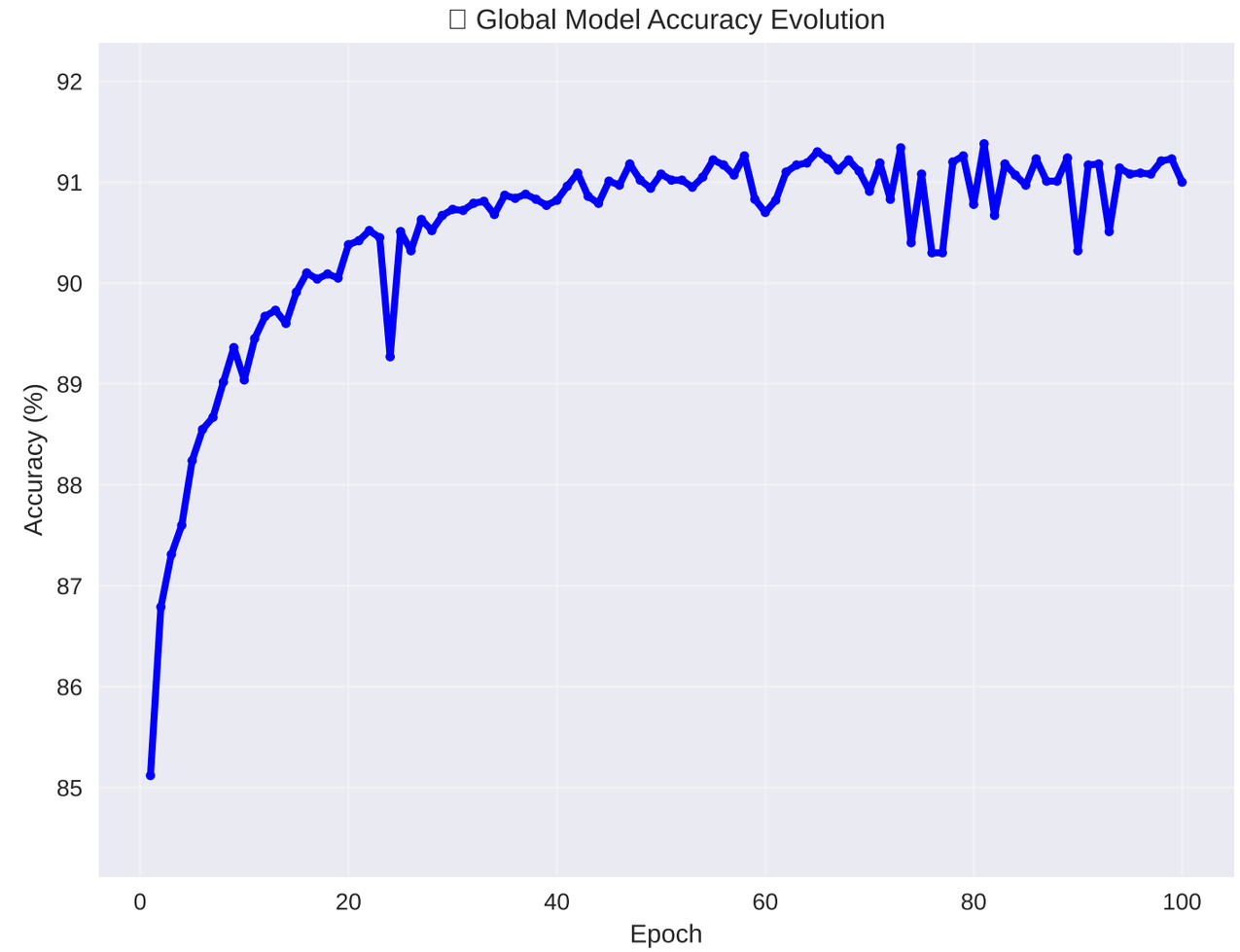
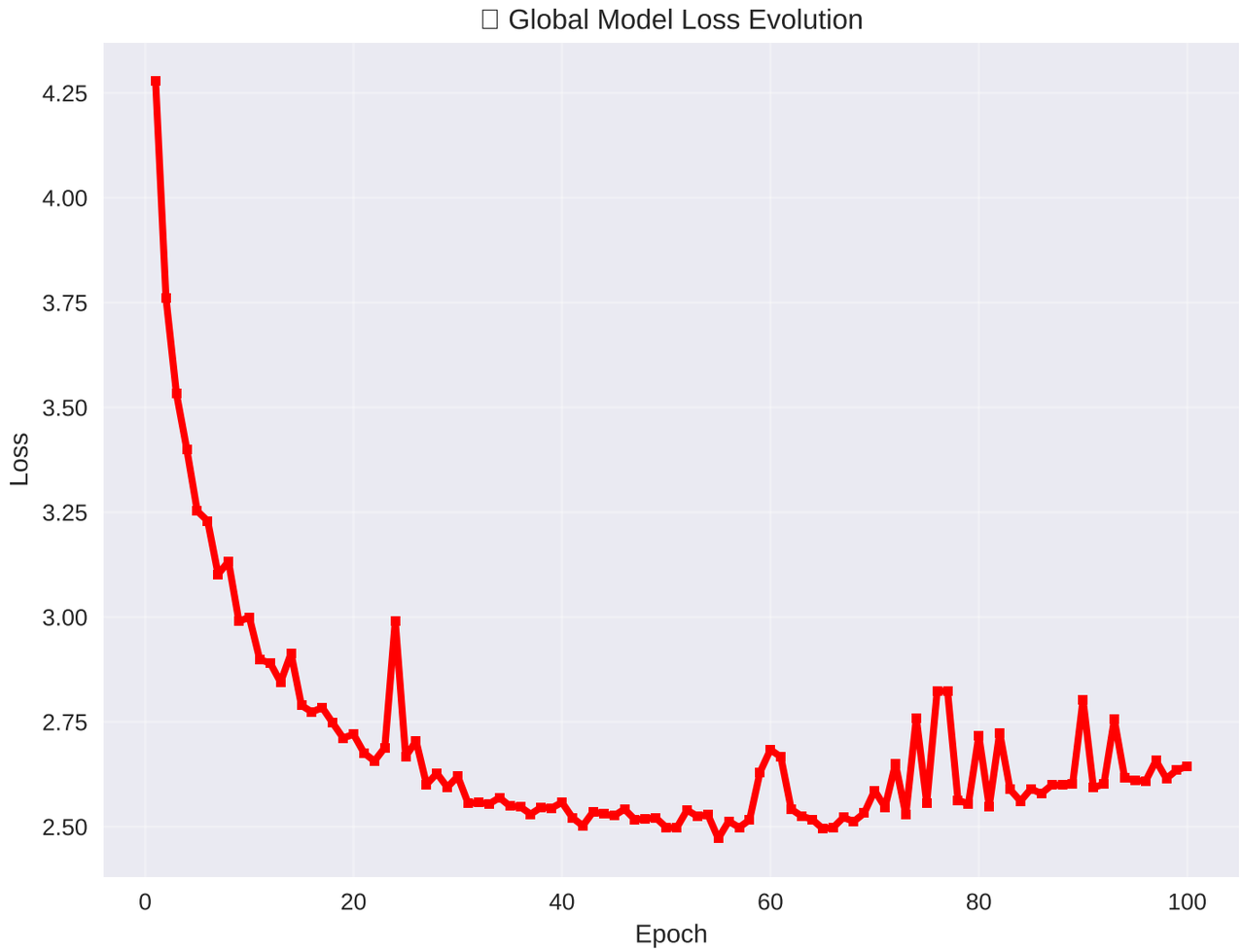
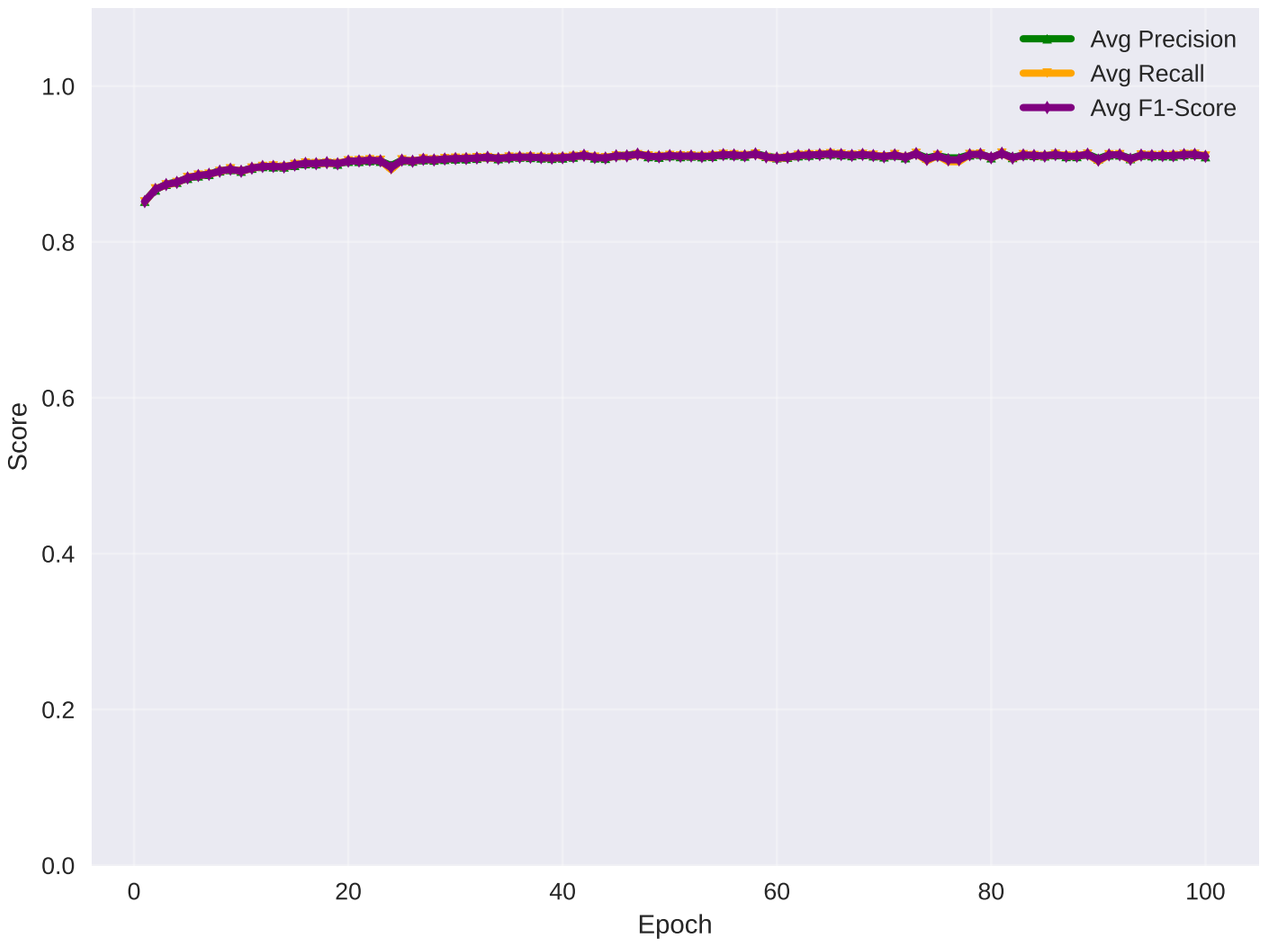
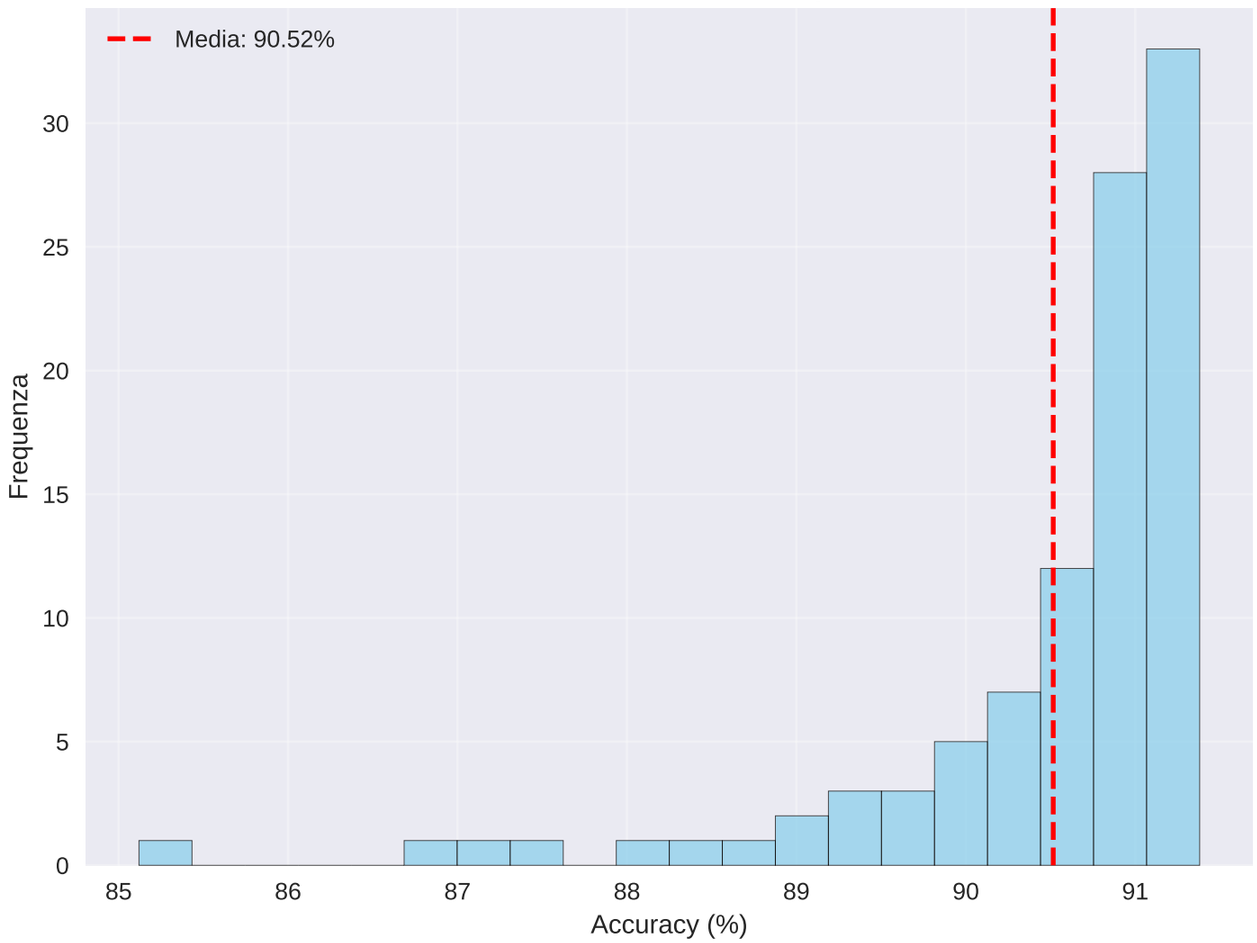
Federated Learning - Main Metrics Overview



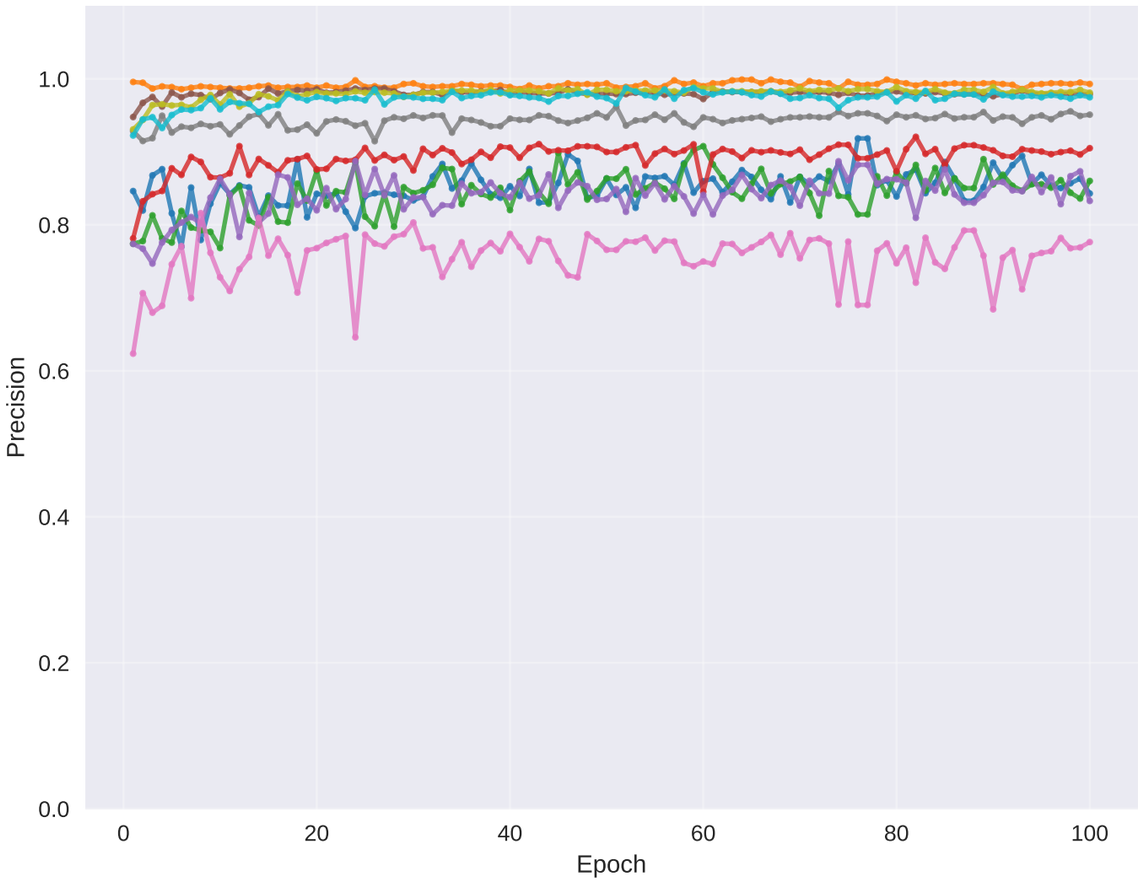




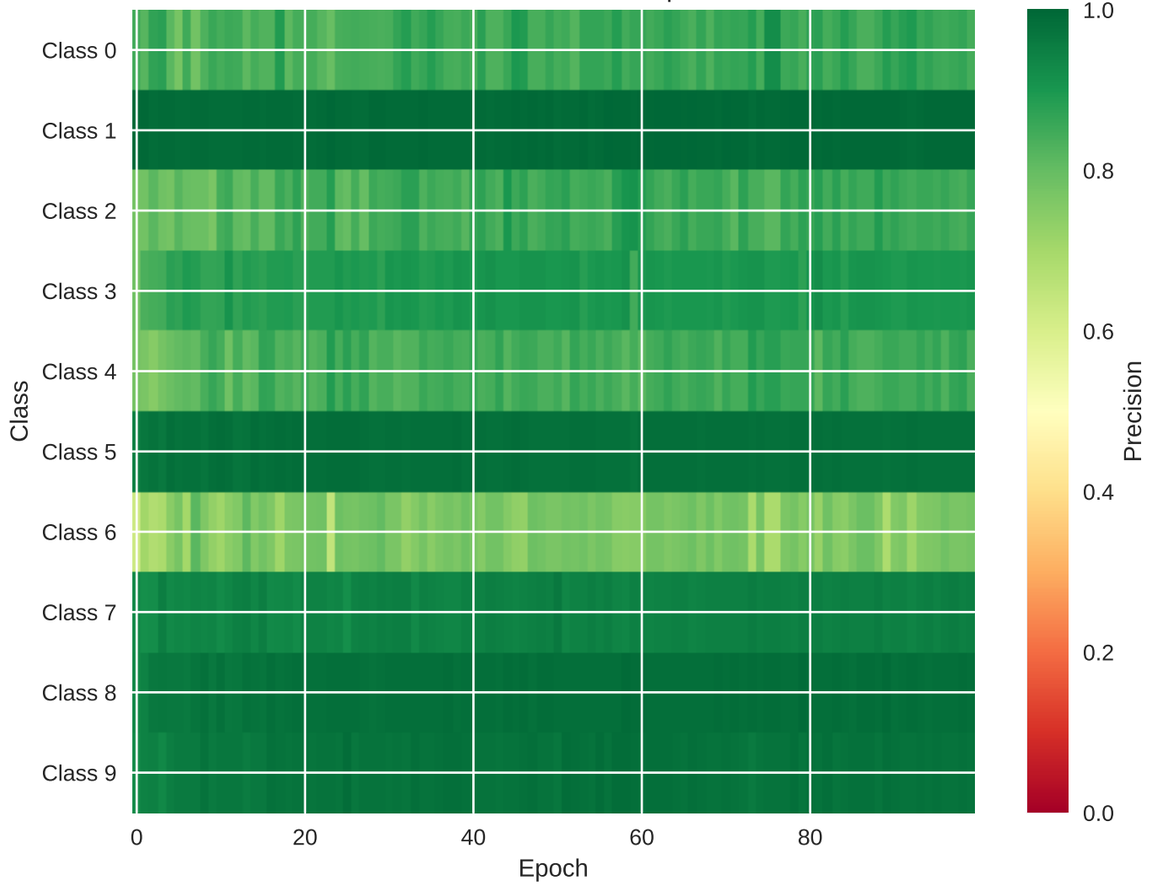


Per-Class Precision Analysis

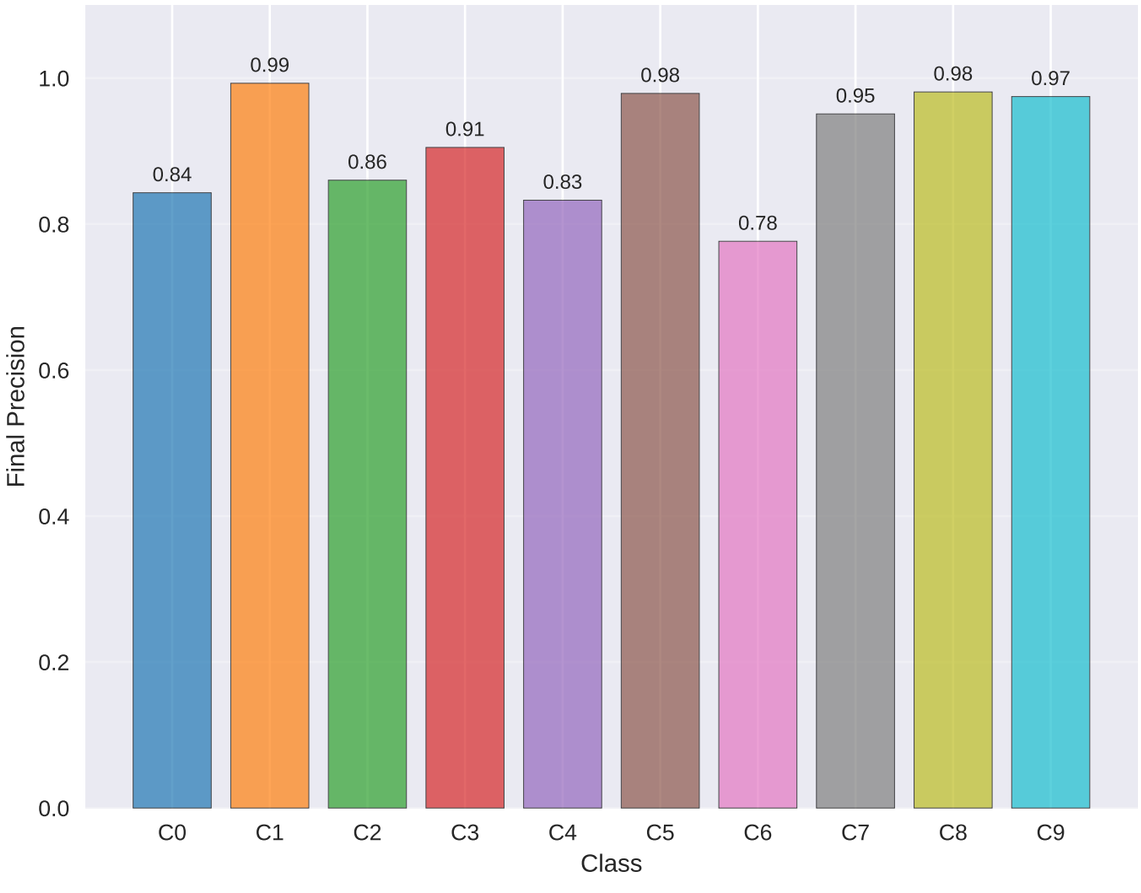
□ Per-Class Precision Evolution



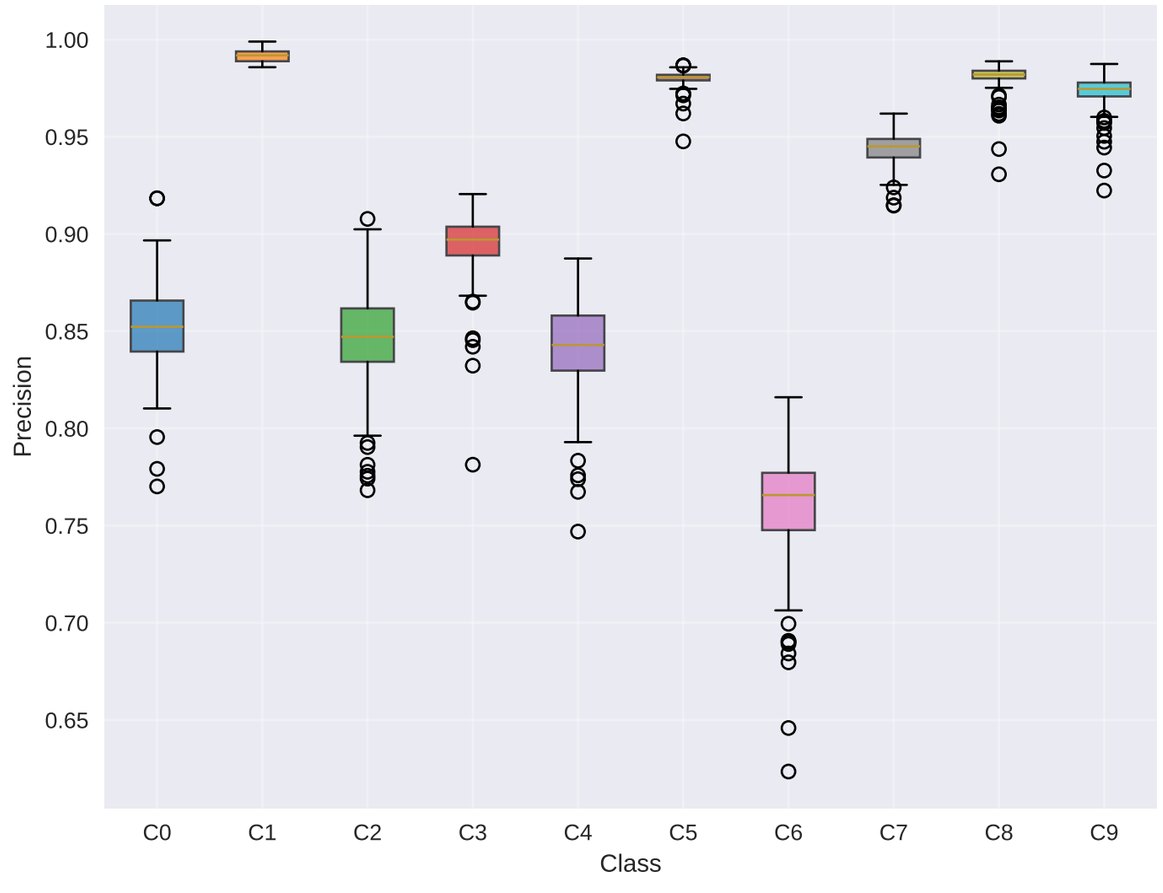
□ Precision Heatmap



□ Final Precision per Class

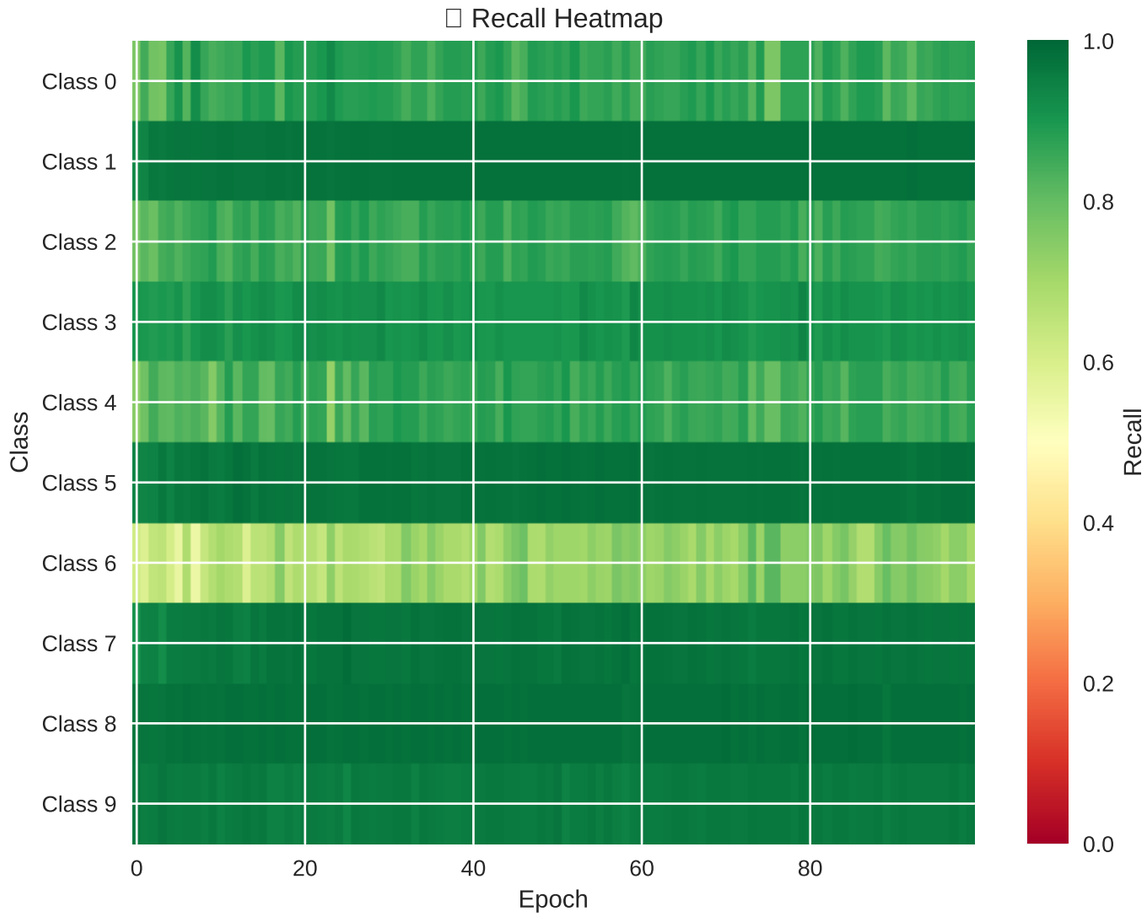


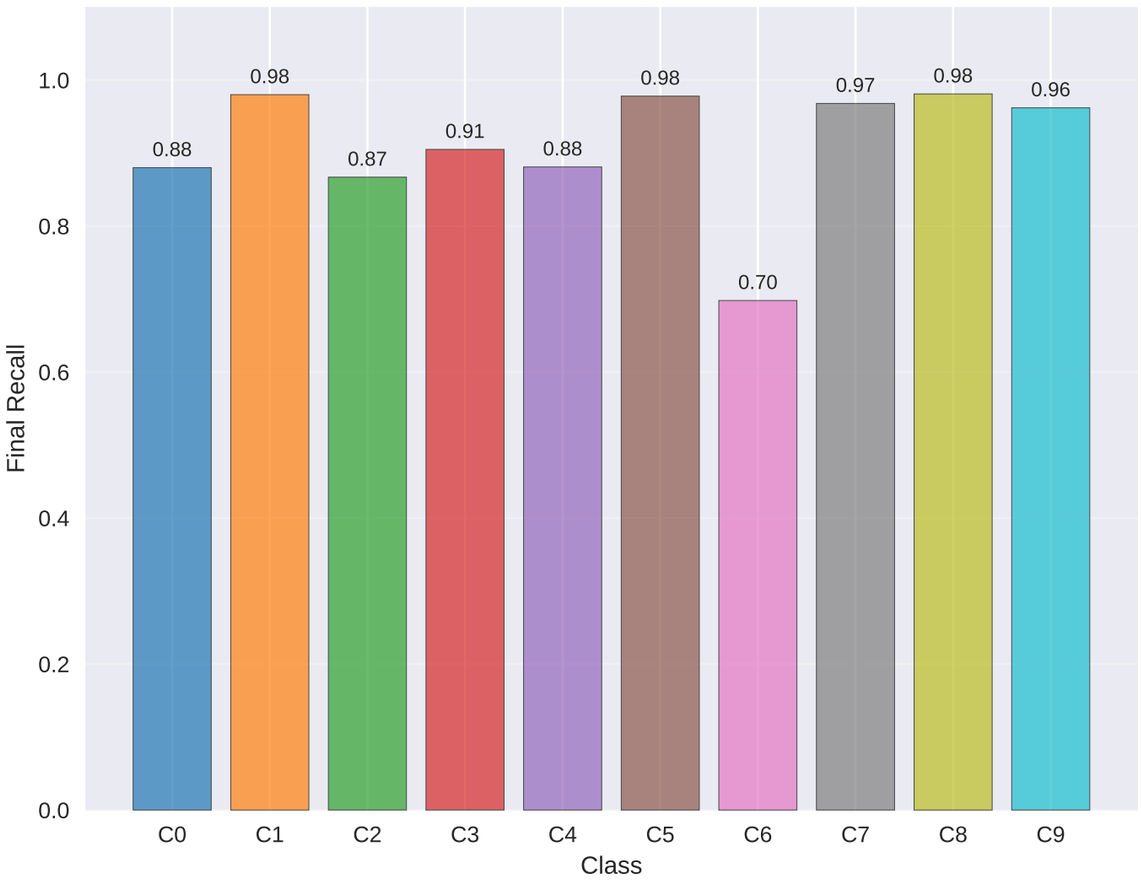
□ Precision Distribution per Class

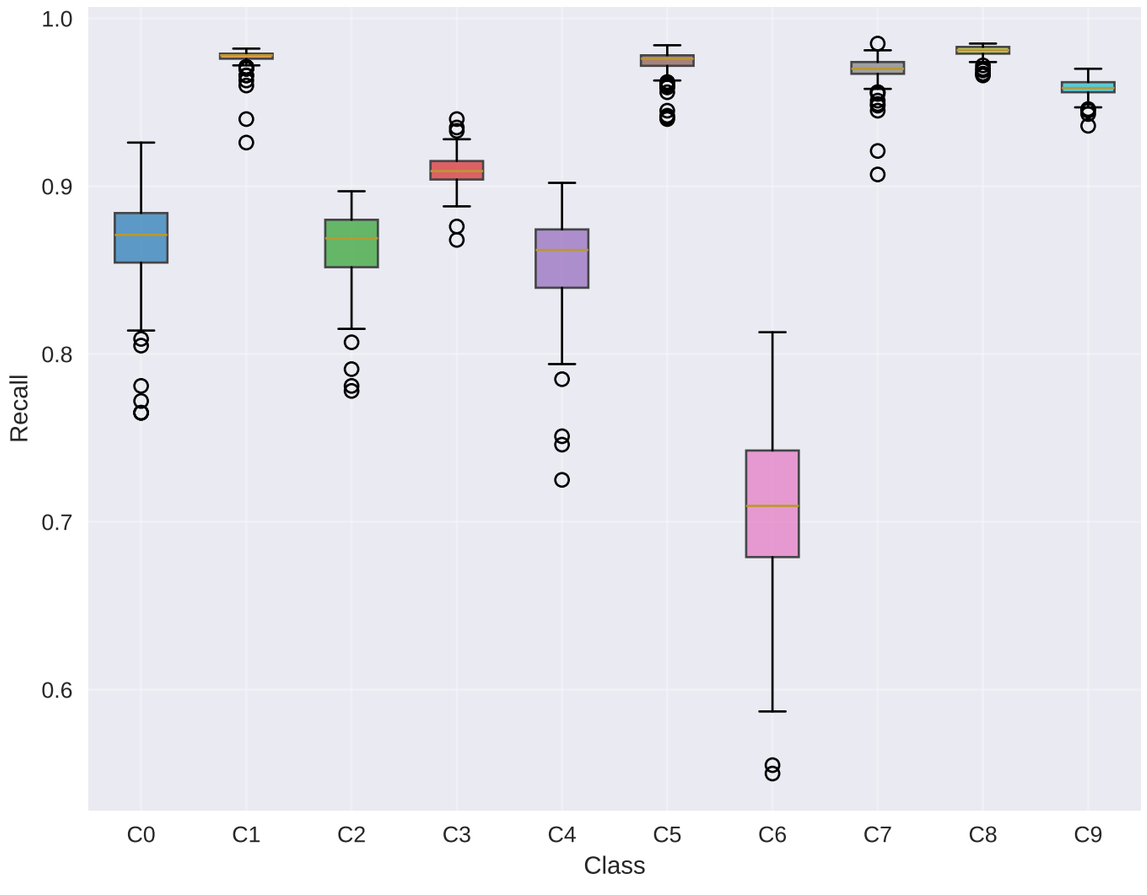


Per-Class Recall Analysis

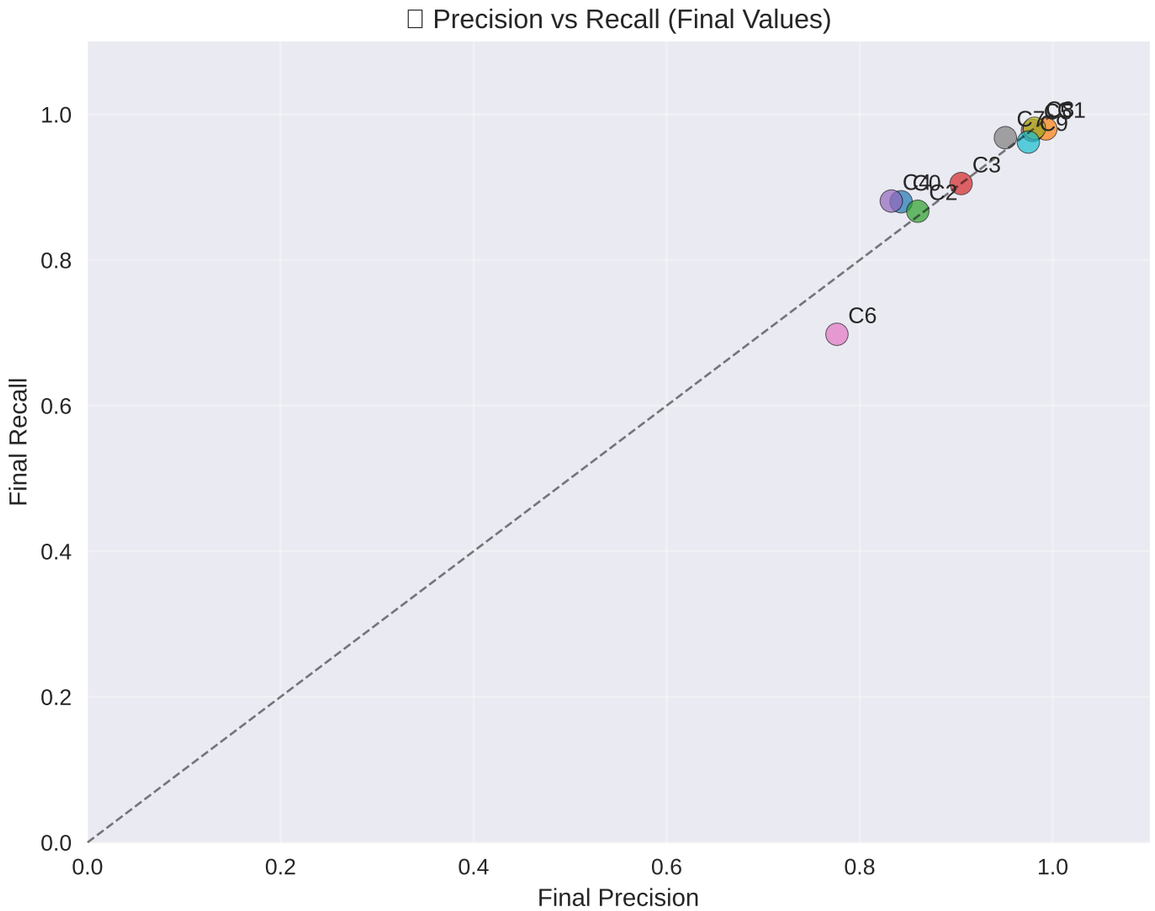




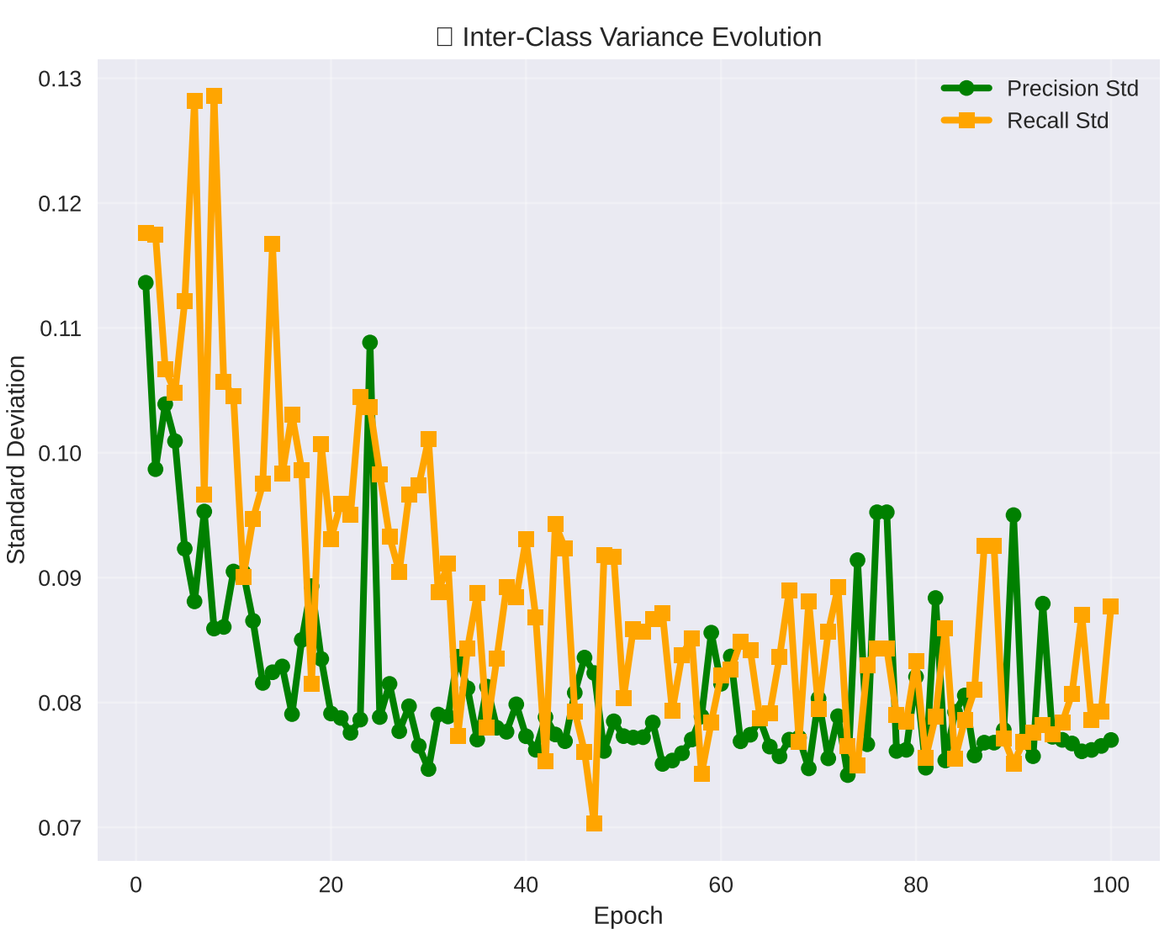


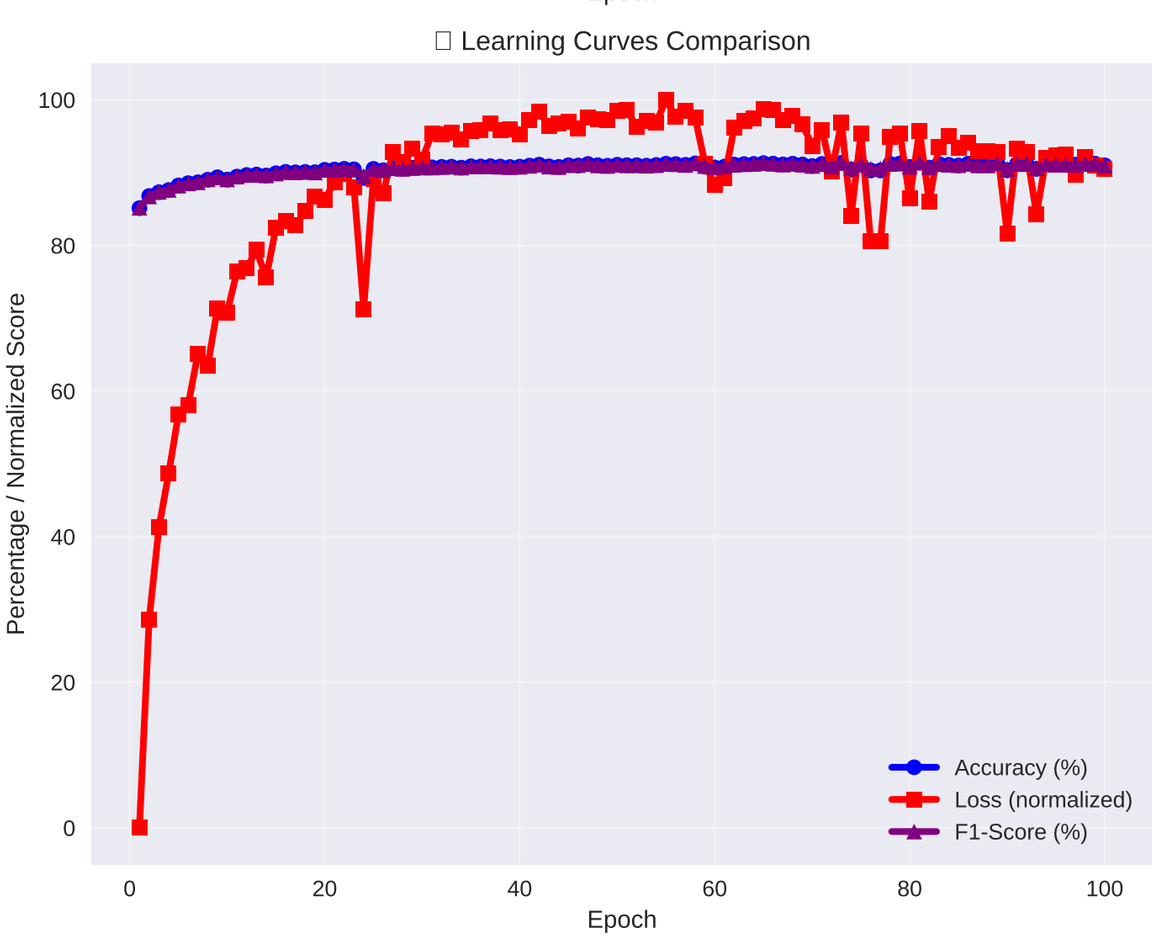


Comparative Analysis









Performance Summary Report

```
FEDERATED LEARNING - PERFORMANCE SUMMARY
========================================

FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-02 22:20:40
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.00%
• Loss: 2.6435
• Average Precision: 0.910
• Average Recall: 0.910
• Average F1-Score: 0.910

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +5.88%
• Riduzione Loss: +1.6354
• Stabilità Accuracy (CV): 0.0114

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.993)
• Peggiore Precision: Classe 6 (0.776)
• Migliore Recall: Classe 8 (0.981)
• Peggiore Recall: Classe 6 (0.698)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.843, R=0.880, F1=0.861
• Classe 1: P=0.993, R=0.980, F1=0.986
• Classe 2: P=0.860, R=0.867, F1=0.864
• Classe 3: P=0.905, R=0.905, F1=0.905
• Classe 4: P=0.833, R=0.881, F1=0.856
• Classe 5: P=0.979, R=0.978, F1=0.978
• Classe 6: P=0.776, R=0.698, F1=0.735
• Classe 7: P=0.951, R=0.968, F1=0.959
• Classe 8: P=0.981, R=0.981, F1=0.981
• Classe 9: P=0.975, R=0.962, F1=0.968

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
```